When Is History Long Enough? Conditional Quantile Models versus Historical Rules for Peak Net-Position Limits

Abstract

Peak-position limits concern the largest absolute inventory reached before a review ends, making completed path labels and the observed current position central to forecasting. We compare historical and conditional quantile reports for an FX market maker and weekly primary-dealer position aggregates, using matched histories and update schedules. Simple historical rules remain demanding benchmarks across frequencies, position categories and reference choices. At the shorter horizon, long-history FX regression lowers matched loss by 14.8% and 16.6% in the previously reported later and earlier windows. A post hoc history-to-horizon display summarizes pooled simulation tasks, and a new simulation passes its prespecified history-region screens. Retrospective FX curves partly agree with the resulting cutpoint bands, while censoring, large-history reversals and failed universal simplification limit that evidence. For limit managers, these findings motivate matching completed-label histories before assessing conditional forecasts jointly on score, breaches and stability.

1 Introduction

An inventory limit can look adequate at the horizon endpoint even when a dealer's position has crossed it earlier in the review period. Figure 1 shows why the relevant forecast is a sampled path maximum rather than an endpoint interval. The challenge is to learn that peak from completed historical labels while preserving the information available when the position is observed. We ask when additional conditioning improves on historical headroom forecasts that use the same training history, and how longer histories alter that comparison.

Simple historical rules provide a demanding reference because they already incorporate information about the current position and its recent path behavior. A conditional model must improve on that information while estimating additional relationships from a limited and changing sample of past peaks. More predictors can help distinguish current conditions, but they also increase sensitivity to the historical sample and the strength of regularization. We examine these competing effects through matched histories, alternative reference rules, and repeated chronological comparisons.

Training-history length provides an organizing question for the comparisons, with the proposed explanation tested rather than imposed on every favorable and adverse result. A post hoc coordinate comparison motivates predictions for a new simulation grid and fixed-period FX learning curves under unchanged validation and evaluation clocks. The resulting support is partial: the broad simulated loss pattern survives, while universal simplification fails and the financial curves show censoring and reversals. This evidence guides history-sensitivity validation without identifying a unique estimation-cost channel or a permanent sample-size threshold.

We study an hourly FX market-maker record and weekly U.S. primary-dealer position aggregates, asking where conditioning improves peak forecasts. In the original ordered comparisons, historical rules, regression and local quantiles compete for the same peak using issue-known observations. These comparisons reveal whether gains persist across reference rules and periods, and whether better scores also mean fewer breaches.

The clearest repeated gains arise for long-history FX regression at the shorter forecast horizon, rather than across every conditional specification. Against a historical headroom rule with the same history, it lowers later-period loss by 14.8%. The earlier-period reduction is 16.6%, and the delayed coverage adjustment preserves the later advantage. Changing the comparison to the shorter-history rolling reference reverses the direction of the earlier-period result, while weekly comparisons frequently favor historical rules.

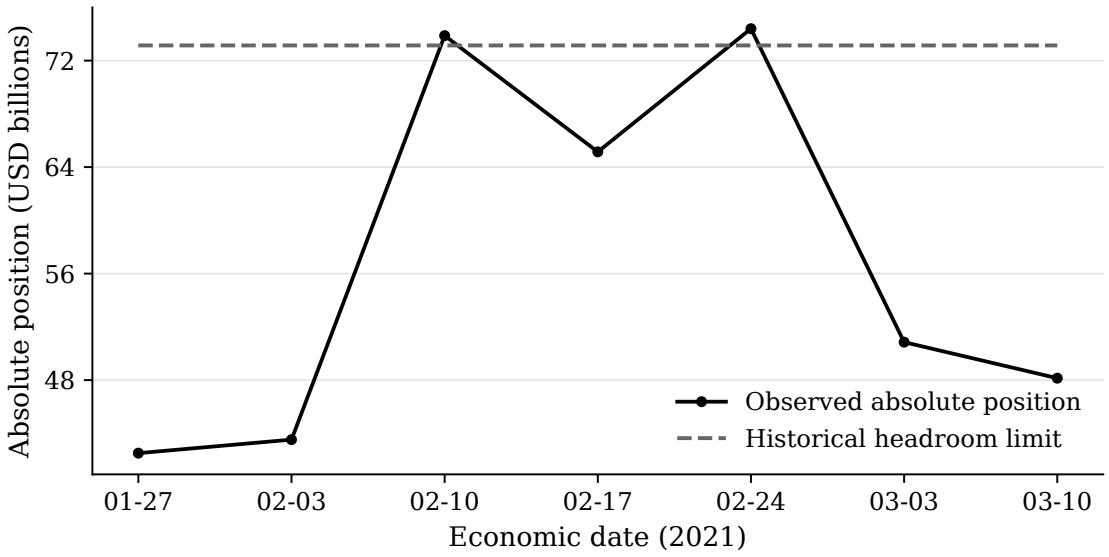


Figure 1: A selected Treasury-bill path whose endpoint returns within the historical report after an intervening breach. The rule chooses the earliest eligible original evaluation window. Endpoint containment can conceal a peak breach.

A lower quantile score also leaves a separate operational question about how often the predicted limit is crossed during the forecast period. The later matched FX comparison records 376 regression breaches, compared with 360 for its historical reference. For the separately named comparisons, Murphy diagrams plot mean elementary quantile scores across thresholds, revealing where consistent scoring rules give different rankings. Error-weight comparisons show how preferences between fixed forecasts depend on the relative penalties on underprediction and overprediction. For managers setting monitoring limits, the practical implication is to evaluate prediction errors and exceedance frequency together before selecting a more complex forecast.

C1. We formalize the peak-position quantile as a path-dependent forecast with a known current-position floor. We record an elementary loss-improving floor projection and the true-quantile strict-tail bound, then apply the classical Lévy inequality under independent symmetric increments. Separately fitted short-history FX regression forecasts cross across horizons on 640 evaluation origins. Wisniewski et al. (2020) forecast signed endpoint intervals rather than a sampled absolute inventory peak. The portfolio-loss maximum of Boudoukh et al. (2004) and payment-deficit peak of van den Hoogen (2017) provide distinct path-risk precedents. Section 3 provides the definitions, propositions and crossing diagnostic, with proofs and simulation qualifications in the Supplement.

C2. We describe history-dependent conditional gains on a post hoc history-to-horizon display, with partial support for specified history regions from a new simulation grid and retrospective FX curves. We compare the same peak target for one FX record and thirteen weekly position-category aggregates with matched histories and update schedules. Even after extending the regression penalty menu, historical rules win in 34 of 48 additional-panel comparisons. The new simulations satisfy their short-history and long-history criteria, while the predicted universal advantage of the smaller regression fails in the comparable cells. Where Duffie et al. (2023) examine capacity, illiquidity and PnL quantiles, and Wisniewski et al. (2020) forecast one market maker's endpoints, we compare these peak-position tasks. Section 7 and Figure 4 connect that contrast to post hoc first cutpoints, left censoring and long-history reversals, guiding matched history-sensitivity validation.

C3. We show how peak-limit rankings change with the historical reference and breach assessment in these observed forecasting tasks. We compare same-history and shorter-history rolling controls across three windows with previously examined outcomes, using established score diagrams and error-weight preferences. For the distinct long-horizon legacy comparison, the preferred report reverses at a penalty ratio of about 20.6:1. Where the endpoint lineage evaluates intervals and Ehm et al. (2016) develop general score comparisons, we connect these diagnostics to monitoring limits with completed path labels. Sections 6–8 support a four-part checklist: use matured labels, separate history and model changes, report scores with breaches, and check stability across periods and related series.

The remaining sections define the peak target, describe the data and forecasts, and compare accuracy, breaches and stability before drawing implications for model validation. Complexity earns its place through a stable comparative advantage on the quantity that the limit actually monitors.

2 Related literature

Net-position forecasting already has a close empirical lineage in the market-making literature, where Wisniewski and colleagues predict signed next-hour endpoint intervals. Their methods combine historical positions, calendar information, neighboring observations, and normalized prediction intervals, with sequential training, calibration, and evaluation (Wisniewski et al., 2020; Wisniewski, 2024). They jointly report scores and coverage diagnostics, including empirical conditional-coverage failures, providing a direct precedent for evaluating accuracy with hit behavior. Our response instead requires the complete future path segment and includes the current absolute position as a known lower bound. This distinction creates different training labels and different limits, even when the numerical forecasting engine follows the same broad family. Comparing their paired endpoint scores with our upper-peak score requires refitting the competing engines to common completed peak labels.

Financial path-risk studies document that a within-period extreme can matter more than the value observed at the end of the period. MaxVaR evaluates the worst cumulative portfolio return under a fixed portfolio and diffusion assumptions, including a conversion from endpoint probabilities to path risk (Boudoukh et al., 2004).

The payment-liquidity thesis constructs a conditional quantile buffer for the maximum cumulative negative payment balance using transaction-path simulation and extreme-value tail estimation (van den Hoogen, 2017). Its cumulative cash deficit differs from sampled absolute inventory, and its confidential records and complete engine remain outside our executed comparisons. Related settlement studies examine tail liquidity needs and collateral requirements in their respective payment systems (Petrunia et al., 2019; Arjani et al., 2020). Duffie and colleagues analyze dealer capacity and Treasury-market functioning, with additional out-of-sample PnL quantile comparisons in their appendix (Duffie et al., 2023). Our empirical contribution concerns matched forecasts of the present peak-position target across the FX record and related public position aggregates.

Quantile estimation and evaluation supply established tools for comparing conditional peak forecasts against historical rules on a shared response. Regression quantiles target asymmetric forecast errors, providing the basic statistical building block for the proposed conditional reports (Koenker & Bassett, 1978). Conditional autoregressive quantiles and quantile forests offer additional conditioning structures for forecasting applications that differ in response, predictors and estimation (Engle & Manganelli, 2004; Meinshausen, 2006). Learning-curve research documents diverse and sometimes nonmonotone relationships between training size and predictive performance across methods and data conditions (Viering & Loog, 2023). Our fixed-period curves

compare relative loss against a historical reference that also changes with training history, differing from an independent-sample expected-risk curve. Consistent scoring tools explain ranking differences across evaluation questions (Gneiting, 2011; Ehm et al., 2016); Table 1 locates the shared peak target, mature labels and matched reference design.

Table 1: Closest literature and the present forecasting task. n.r. marks a dimension unreported in the consulted task or passage. The distinction is a common peak-position target and its validation design.

Study	Response	Data unit	Peak floor	Label timing	Evaluation	Source scope
Wisniewski et al. (2020)	Signed endpoint interval	One FX market maker	n.r.	Ordered train/ calibrate/ test	Scores and hit tests	One source family
Boudoukh et al. (2004)	Worst portfolio return	Stylized fixed portfolio	Different initial-value anchor	Analytic diffusion	Tail ratios and monitoring	Parameter cases
van den Hoogen (2017)	Cumulative payment-deficit peak	Confidential bank payments	Absolute-position floor n.r.	Path simulation	EVT methodology; results partly confidential	Bank payment task
Duffie et al. (2023)	Capacity, illiquidity, PnL quantiles	Dealer/ market sources	n.r.	Expanding PnL fits	Capacity and PnL forecasts	Several sources
Engle and Manganelli (2004)	Return quantile	Asset returns	n.r.	Lagged predictors	Quantile scores and DQ	Several returns
Ehm et al. (2016)	Quantile/ expectile comparison	Multiple forecast examples	n.r.	Original example designs	Elementary-score curves	Multiple tasks
Meinshausen (2006)	Conditional response quantile	Regression examples	n.r.	Original regression splits	Conditional distributions	Several datasets
This study	Sampled absolute-position peak	One FX record; weekly totals	Observed current magnitude	Completed forward labels	Scores, breaches, coherence	Related aggregate tasks

The institutional payment-thesis reading is partial; n.r. is not evidence of absence. The earlier FX work and this paper share a source family, without verified input-byte identity.

3 The forecast target

The forecast target is the largest sampled absolute position between the issue time and the end of the stated forecast horizon. Let N_t denote the reported signed position and let $\mathcal{F}_t$ contain the information available at the forecast origin. Define the peak $Y_{t,H}$, its observed floor B_t, and the corresponding nonnegative headroom $Z_{t,H}$ by

$$Y_{t,H} = \max_{0 \le h \le H} |N_{t+h}|, \qquad B_t = |N_t|, \qquad Z_{t,H} = Y_{t,H} - B_t \ge 0.$$

The inclusion of the issue position matters because a feasible peak limit must already cover the exposure observed at the forecast origin. Figure 1 shows a real window selected by a fixed chronological rule, with an endpoint inside the limit and an earlier peak outside it. A successful endpoint interval can therefore answer a different risk question from a successful report for the full sampled peak. We evaluate the latter through pinball loss and count a breach precisely when the observed peak strictly exceeds the reported limit. This report monitors sampled net inventory; its known floor neither measures settlement cash needs nor establishes an optimal operating limit.

$$\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+, \qquad I_{t,H} = \mathbf{1}\{Y_{t,H} > C_{t,H}\}.$$

Proposition 1(known-floor projection). Suppose $Y \geq B$ almost surely and the issue information contains B. Projecting any forecast c to $\max\{B, c\}$ weakly improves pinball loss for every possible outcome, with an exact gain below the floor. If q is the true conditional lower inverse quantile at level $1-\alpha$, its strict exceedance probability is at most α. At an atom of the response distribution, that probability can be smaller than the nominal tail probability.

$$\rho_\tau(Y-c) - \rho_\tau(Y - \max\{B,c\}) = \tau(B-c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \leq \alpha.$$

Proposition 2(classical endpoint-to-peak bound). Let future increments be independent and individually symmetric about zero, with partial sums S_h. Fix a starting position x, set $B = |x|$, and let q be the population $(1-\alpha/2)$ quantile of $|S_H|$. The triangle inequality and the classical scalar Lévy maximal inequality then imply the following bound, including strict inequalities at atoms (Sussman, 2023). Neither identical increment distributions nor finite increment moments are required for this population statement.

$$\Pr\left(\max_{0\leq h\leq H} |x + S_h| > B + q\right) \leq \Pr\left(\max_h |S_h| > q\right) \leq 2\Pr(|S_H| > q) \leq \alpha.$$

The Supplement gives both proofs and distinguishes the population endpoint quantile from the empirical endpoint estimate used by the corresponding benchmark. A conditional use of Proposition 2 requires the independence, symmetry, and endpoint-quantile assumptions under the law conditional on issue information. Its simulation comparisons separate mean-reverting position levels from serially correlated increments, since those constructions have different dependence in their future paths. This separation makes the mathematical premises explicit before the empirical benchmark is judged by its observed losses and breaches.

At the same issue origin and quantile level, the conditional lower-inverse quantiles of nested peaks are ordered. Separate fitted reports can cross, as can differing selections from a set-valued quantile functional. The short-history FX regression reports cross across horizons on 640 origins, illustrating a coherence problem distinct from the score of each marginal forecast. For equal horizon weights and a common quantile level, sorting crossed reports weakly reduces their combined pinball loss for nested realized peaks. This aggregate property has separate scope from improvements at each horizon or in coverage. The analytic qualification in the diagnostic addendum gives the elementary identity and corrects the older Supplement's broader repair wording.

4 Data

The FX application uses the hourly member of the publicly described Net Position Time Series Dataset, representing a market maker's net client-flow position (Lindsay, 2020; Wisniewski et al.,

2020). The accepted source contains 18576 records, with positions measured in the source's USD units and ordered by its observation clock. We use the same series for the later evaluation and the earlier chronological comparison, with their separate evaluation indices preserved throughout. Within each comparison, every candidate forecast is scored against the same sampled peak and the same current-position floor. The nominal hourly records retain weekend gaps, so a fixed record horizon can span different stored-clock durations.

The weekly application uses Primary Dealer Statistics published by the Federal Reserve Bank of New York (Federal Reserve Bank of New York, n.d.). Its reporting definitions specify fair-value outright positions, aggregated as long minus short, under the government-securities dealer framework (Board of Governors of the Federal Reserve System, 2013, 2024). The original Treasury-bill aggregate has 706 observations, with values expressed in USD millions and indexed by their economic dates. Additional position categories use the same complete weekly date grid after a metadata and calendar eligibility check conducted before their model evaluation. These categories cover different reported position components and are fitted separately rather than pooled into a common monetary forecasting sample.

The FX record describes one market maker, while the weekly series combine signed positions across primary dealers. The weekly experiment replays a fixed downloaded vintage on economic-date indices. It stipulates that current and earlier snapshot values are known at each issue, making the current-position floor known under that replay convention. Public release occurs later; these inputs therefore define an idealized information set rather than a historical public observer's information (Federal Reserve Bank of New York, n.d.).

The reporting definitions identify the positions being aggregated. Each analysis uses its own units, training history, forecast horizon, and issue grid, with reference rules matched inside the relevant comparison. The data design makes frequency and measurement differences visible instead of treating every sequence as an interchangeable dealer history.

Daily aggregation of the same dealer produces 930 nonempty source-naive calendar days and 501 common evaluation issues. Each day contributes its last record, including partial boundary days, and the report clock follows the end of that day. The timezone and historical release vintage remain unknown; this last-record proxy is distinct from a verified market close or intraday maximum.

5 Forecasting methods

The primary forecasts target the conditional quantile at $\tau = 19/20$, with its upper tail defining the strict-breach event. Its nineteen-to-one error preference is hypothetical and is separate from measured funding, trading or settlement costs. The two horizons span $H \in \{6, 24\}$ subsequent source records, and each peak also includes the position observed at issue time. This common target keeps the comparisons aligned while allowing the historical pool and conditioning structure to differ between the named procedures.

Historical rules serve as substantive competitors because they can already adapt their limits to the position observed at the forecast origin. The historical headroom rule adds an empirical quantile of completed past headroom to the current absolute position, while the translated-path rule reanchors historical signed paths. The historical-peak rule uses completed absolute peaks directly, and the endpoint-derived rule adds an empirical absolute-displacement quantile to the observed floor. Table 2 gives their definitions, update schedules, and the parameter menus used by the conditional methods.

Conditional quantile regression links completed peak headroom to issue-known position histories and calendar information, using training-based transformations and ridge regularization. The

original predictor set contains 33 features before its intercept, with all transformations fitted inside the chronological training sample. The parsimonious variant concentrates on a smaller state description, while the expanded ridge comparison changes the penalty menu with the richer predictor set held fixed. These comparisons distinguish changes in the conditioning information from changes in the strength of regularization rather than attributing every gain to feature reduction.

Local quantile methods condition on neighboring historical states, with a scale-adjusted variant using training information to normalize the comparison. The filtered-path benchmark recursively simulates signed positions using GARCH-standardized empirical innovations drawn independently with replacement (Hull & White, 1998; Barone-Adesi et al., 1999). The other benchmarks estimate a quantile from response distributions in tree leaves or fit tails of volatility-scaled peak headroom (Meinshausen, 2006; McNeil & Frey, 2000). These are the existing fitted specifications, with every complete result and numerical or structural failure documented in the Supplement. Their different conditioning assumptions make the actual executed procedure, rather than its family name alone, the relevant object for comparison.

The further robustness comparisons address calibration, forecast combination, and conservative selection while keeping the peak response and chronological information restrictions explicit. A headroom version of the symmetric absolute-value quantile recursion follows the CAViaR family, and finite delayed calibration wrappers update only after outcomes mature (Engle & Manganelli, 2004; Gibbs & Candès, 2021). Fixed mean and median combinations aggregate the specified forecasts, while a paired-difference one-standard-error rule favors simpler reports within its stated validation tolerance. Every method earns its comparison against the historical headroom rule through its actual score and breach outcomes on the shared evaluation origins.

Table 2: Forecast definitions and executed menus. Codes occur here to connect with the result files. Matched histories and update schedules make the reference explicit.

Method / code	Forecast definition	Executed menu / setting
Historical headroom (F)	Current floor plus a quantile of completed peak headrooms	FX histories {720,2880}; weekly {52,260}
Translated path (S)	Translate completed signed paths to the current position before taking their peaks	Same completed-label pools; fixed-window and rolling-window controls
Historical peak (K)	Historical absolute-peak quantile projected to the current floor	Same histories and update controls
Quantile regression (QR)	Training-scaled headroom regression with ridge slopes and an unpenalized intercept	Original penalties $\{0.001, 0.01, 0.1\}$; expanded menu in Supplement
Local quantiles (KNN)	Neighbor headroom quantile, with a separate increment-scale variant	Training-only transforms and fixed neighbor menus; Supplement
Parsimonious regression (QR4)	Current signed/magnitude positions and short/long increment RMS	Separate five-point penalty menu; common-menu comparison in Supplement
Endpoint plug-in (L)	Floor plus a high absolute-displacement endpoint quantile	Proposition 2; empirical plug-in evaluated separately
Further benchmarks	Autoregressive replay; filtered paths; peak-tail fit; quantile forest	Existing bounded specifications, complete failures included in Supplement
Coverage and combinations	Delayed projected ACI-style wrappers; headroom SAV; fixed forecast combinations	Original-source wrapper/SAV inventory and broader combination inventory are separate

Training histories count completed labels. Fixed-window models refit every 24 FX records or 13 weekly records. Quantile and normalization conventions are detailed in the Supplement; this table does not expand any executed menu.

6 Evaluation design

The forecasting clock admits a training label only after the entire response horizon has ended, preventing an unfinished peak from entering estimation. At refit origin r, a historical origin i is eligible only when $i + H \leq r$, and every selected window uses the same completed-label convention. Predictors use issue-known observations, and training-only transformations remain fixed until the next scheduled refit rather than learning from future evaluation outcomes. The original chronological comparisons select parameters in the preceding validation segment and evaluate the complete specified issue grid. The expanded backcast instead retains penalties selected later, although each coefficient fit and transformation still uses only mature training labels.

Matched comparisons hold the completed-response history and refit schedule fixed when evaluating the regression against the historical headroom rule. The original FX primary instead uses the shorter-history rolling historical reference, making it a distinct comparison from the same-history benchmark. The Treasury-bill primary compares the longer-history regression with the historical headroom rule on a shared history and update schedule. Tables 3 and 4 preserve these original comparisons alongside the matched, additional-panel, and earlier-period results.

We evaluate forecast accuracy with mean pinball loss and report strict breaches separately, including their frequency and the associated overrun amounts. Paired loss differences use heteroskedasticity and autocorrelation consistent covariance estimates, with the horizon represented in the bandwidth choice (Diebold & Mariano, 1995; Newey & West, 1987). The existing calculations also report the small-sample adjustment and the stated multiple-comparison families, with each family's role identified explicitly (Harvey et al., 1997; Holm, 1979). Coverage diagnostics use a fixed spaced grid and distinguish unconditional coverage from a transition likelihood conditioned on its initial hit (Kupiec, 1995; Christoffersen, 1998).

6.1 Research design and transparency

The original source-specific primaries retain their original roles, outcomes, and reference procedures throughout this paper. The conditional extension followed inspection of the original FX evaluation archive; later calibration, combination, state, quantile-level, and selection analyses are post-result and exploratory. Their designs and execution records preserve the chronology of each addition, including failed candidates and full comparison families rather than selected favorable subsets. The expanded prefix reuses familiar source observations, overlaps the earlier backcast, and follows previous rolling-validation scoring of its target paths. The new extensions were fixed before their own calculations within an already inspected research program. These earlier tests and the writing-stage HAC comparisons remain descriptive rather than independent or prospective confirmation.

The history-coordinate comparison is a post hoc discovery on an existing simulation grid. We recorded the subsequent simulation equations, random streams, classification criteria and financial crossing predictions before running their new calculations, with local design commits and chronology available in the repository. This is a pre-run specification within an already examined research program, not a public external registration or an untouched financial holdout. The new simulations change innovation laws and random streams, whereas the FX learning curve reuses the known validation and evaluation period.

Every history shares the original validation segment and evaluation start, with complete mature-label pools and the original update convention; unavailable cases would remain NA. Both W2880 cases exactly reproduced all available old scientific arrays, summary fields and parameter selections before other histories were attempted. Some older parsimonious trace fields were absent,

so their unavailable historical comparisons remain outside that claim. The bootstrap holds the selected coordinate fixed when resampling repetition clusters; its endpoint envelope conditions on that choice and the discovery grid. The new paired-loss diagnostics use the original Bartlett HAC convention with a normal reference and no family adjustment, treating financial dependence conditions as working premises.

The Supplement provides the complete prior and later comparison inventories, the exact evaluation clocks, numerical checks, and all incomplete or unsuccessful procedures. We report sensitivity to bandwidth, adjustment family, target level, and state definition as separate diagnostic questions within their specified scopes. The subsequent interpretation follows the same sequence: specify the reference and sample, compare losses, examine breaches, and assess stability across periods. A credible peak-limit gain must survive this sequence with its benchmark and its information set stated clearly.

7 Results

7.1 Where conditional models improve

Long-history FX regression provides the clearest repeated improvement against a historical headroom rule using the same completed history and update schedule. It lowers matched loss by 14.8% in the later period and 16.6% in the earlier period. The same-history comparison fixes the rolling pool of completed fitting labels and the refit schedule. Feature construction and penalty selection retain procedure-specific raw lookback and validation information. The shorter-history rolling reference also changes the fitting-label history. The matched gains in Table 3 therefore isolate a different comparison from the original rolling-reference result.

The delayed ACI adjustment preserves the matched FX score advantage while bringing the two observed breach frequencies close together. Adjusted regression lowers mean loss by 15.0% against the adjusted historical reference on their common evaluation grid. Their strict breach counts are 329 and 328, respectively. This comparison preserves a relative score advantage, with nearly equal breach frequencies. Each wrapper has higher mean loss than its own unwrapped base, so preserved relative advantage does not mean absolute score improvement from calibration.

The long-history FX gain also persists across the evaluated position-increment states and at the lower target quantile. The unchanged primary has a negative mean loss difference in every training-defined RMS group, with the group results documented in the Supplement. At the lower target, regression lowers loss by 10.1% against matched headroom and by 9.2% against the original rolling reference. These results broaden the description of that configuration while leaving the separate upper-target validation failures visible in the full record.

7.2 Where conditional models lose

The advantage changes when the reference changes, even for the same long-history FX regression report at the shorter horizon. Against the original shorter-history rolling rule, it lowers later-period loss by 12.7% but raises earlier-period loss by 3.2%. The earlier matched improvement therefore coexists with an adverse original-primary comparison, rather than reproducing the original gain against every historical rule. The reversal in Table 3 shows that a matched-history gain can coexist with failure against the simpler operational reference.

Short histories and longer horizons produce additional adverse outcomes that prevent a broad ranking of regression above historical forecasting rules. The short-history FX regression at the longer horizon raises mean loss by 31.5% relative to its matched historical headroom rule. Its

Table 3: Long-history short-horizon FX regression under two references and periods. Negative percentages favor regression. Matched gains recur while the rolling-reference direction changes.

Period / procedure	Reference	Loss change (%)	Issues	Strict breaches (regression / reference)
Later / original primary	Shorter-history rolling headroom	-12.66	6625	376 / 332
Later / matched	Same-history fixed-window headroom	-14.77	6625	376 / 360
Earlier / original structure	Shorter-history rolling headroom	+3.17	1440	See complete earlier-period tables
Earlier / matched	Same-history fixed-window headroom	-16.62	1440	20 / 16
Later / coverage-wrapped	Same-history wrapped headroom	-15.00	6625	329 / 328

Regression uses history 2880 and horizon 6; the original rolling reference uses 720. The earlier matched comparison is the original three-penalty regression, not the later one-standard-error selection rule. The wrapper comparison is a separate finite projected delayed ACI-style exercise.

breach counts are 924 for regression and 427 for the reference. Other long-horizon comparisons show smaller score gains alongside more breaches, so a positive result at the shorter horizon cannot stand in for the full map.

The original Treasury-bill primary loses on both score and breach count. Its loss increase is 50.8%, with 54 regression breaches on the evaluation grid. The historical reference records 28 breaches, as Table 4 shows together with the corresponding fixed-reference inference. The earlier shorter-history structural analogue also loses, so the weekly evidence challenges transport of the successful FX configuration to these aggregate tasks.

Heavier regularization improves the additional weekly panel substantially but leaves historical rules ahead in most of its evaluated cells. The richer regression's median excess loss falls from 92.6% to 9.2% after expansion of its ridge menu. Even then, it loses 34 of 48 comparisons against matched historical headroom rules. The earlier-period one-standard-error exercise misses its success criteria, reinforcing the need to examine stability rather than selecting a favorable regularization summary.

Simpler conditional descriptions and forecast combinations also produce mixed outcomes rather than a uniform solution to the weekly regression losses. The fixed median combination lowers loss against the bill historical reference by 4.0%. The fixed mean combination instead raises loss by 2.8%, and the headroom quantile recursion also performs worse than that reference. Table 4 and the Supplement preserve these comparisons together with the additional model failures, including every unfavorable completed result.

7.3 Scores and breach behavior answer different questions

The matched FX gain comes with more breaches before calibration, showing why pinball accuracy and exceedance frequency deserve separate reporting. Regression breaches 376 times, compared with 360 for the matched historical headroom rule. Pinball loss measures the sizes of underprediction and overprediction; a strict breach records whether the realized peak exceeds its forecast. Managers comparing monitoring limits therefore need both measures, rather than interpreting a smaller average score as a direct reduction in breach occasions.

Table 4: Weekly evidence across the original primary, added panel and earlier-period comparisons. Counts refer to related tasks rather than independent dealers. Stronger regularization narrows losses while leaving substantial failures.

Comparison set	Procedure	Loss change / median (%)	Higher-loss cells
Original Treasury-bill primary	Original regression, long history	+50.79	Original primary
Earlier Treasury-bill analogue	Short-history structural analogue	+144.30	Different history
Additional panel	Original full regression	+92.63	45 / 48
Additional panel	Four-predictor regression	+9.91	37 / 48
Additional panel	Full regression, expanded penalties	+9.22	34 / 48
Additional panel	Ordinary empirical headroom quantile	+0.50	30 / 48
Earlier weekly groups	Validation-minimum expanded rule	+4.15	26 groups
Earlier weekly groups	One-standard-error rule	+5.37	26 groups

The original bill primary has 54 versus 28 strict breaches among 277 issues. Earlier short-history bill results are a structural analogue, not an exact replay of that primary. The one-standard-error worst excess is +181.39%; all its prewritten success criteria remain unmet. Full panel and selection tables are linked from the Supplement and EXTENDED_RESULTS.md.

Murphy diagrams reveal that forecast rankings can depend on the scoring rule even when the quantile level and reports stay fixed. Figure 2 displays crossing curves for the original FX primary, and Figure 3 presents the distinct Treasury comparison. A separate error-weight analysis finds a preference reversal near 20.6:1 for the long-horizon legacy FX report against the rolling translated-path reference. The short-horizon FX primary and matched comparisons have lower mean overrun and unused allowance together, so that particular reversal belongs to the long-horizon comparison.

Table 5 places the working-reference loss tests alongside the fixed-grid coverage results, preserving their distinct null hypotheses and comparison families. A loss-difference test concerns average relative accuracy, while coverage tests concern the frequency and temporal pattern of forecast breaches. State diagnostics add another comparison between conditional predictive ability and differences across the specified RMS groups (Giacomini & White, 2006). The combined evidence supports a narrow score advantage in the successful FX setting and a broader requirement to assess limits through several operationally relevant dimensions.

Table 5: Accuracy and hit diagnostics for explicitly named reports. Panel A reports paired working-reference tests; Panel B uses the fixed nonoverlap grid for hit tests. A better average score does not imply fewer breaches.

Panel A: paired contrast	DM p	HLN p	Named adjustment / p
FX original primary	2.20e-07	2.31e-07	Two-primary Bonferroni: 4.39e-07
FX matched history	4.47e-09	4.83e-09	Original matched Holm family: 2.86e-07
Treasury original primary	0.00761	0.00939	Original matched Holm family: 0.320

Original matched Holm uses 72 contrasts. The later regrouped matched family is separate; earlier-period writing-stage HAC comparisons have neither a new HLN field nor a new family adjustment.

Panel B: report	Grid hits	Grid issues	Kupiec p	Christoffersen CC p
FX regression	50	947	0.695	0.636
FX original rolling reference	49	947	0.807	0.905
FX matched fixed reference	49	947	0.807	0.646
Treasury regression	7	40	0.00413	0.0136
Treasury historical reference	4	40	0.199	0.262

Hit tests use the full fixed nonoverlap grid; the conditional-coverage likelihood conditions on its initial hit. Equality to the nominal tail rate is the stated diagnostic null, distinct from the population upper-quantile inequality. The grid does not create independence.

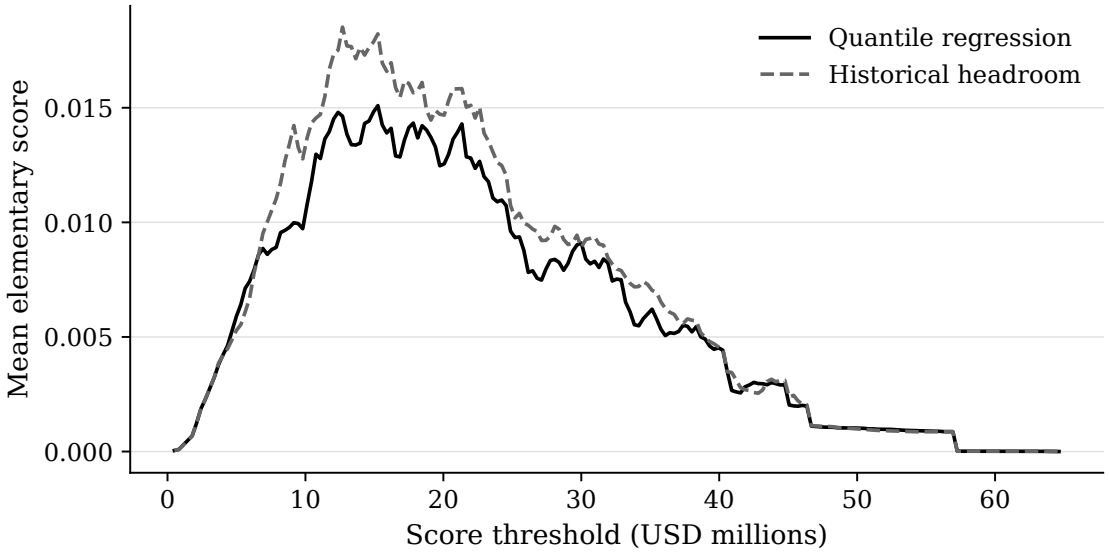


Figure 2: Murphy diagram for the original short-horizon FX primary. The full-breakpoint calculation finds crossings; the graph is a display of saved scores. Average pinball superiority does not imply dominance across elementary scores.

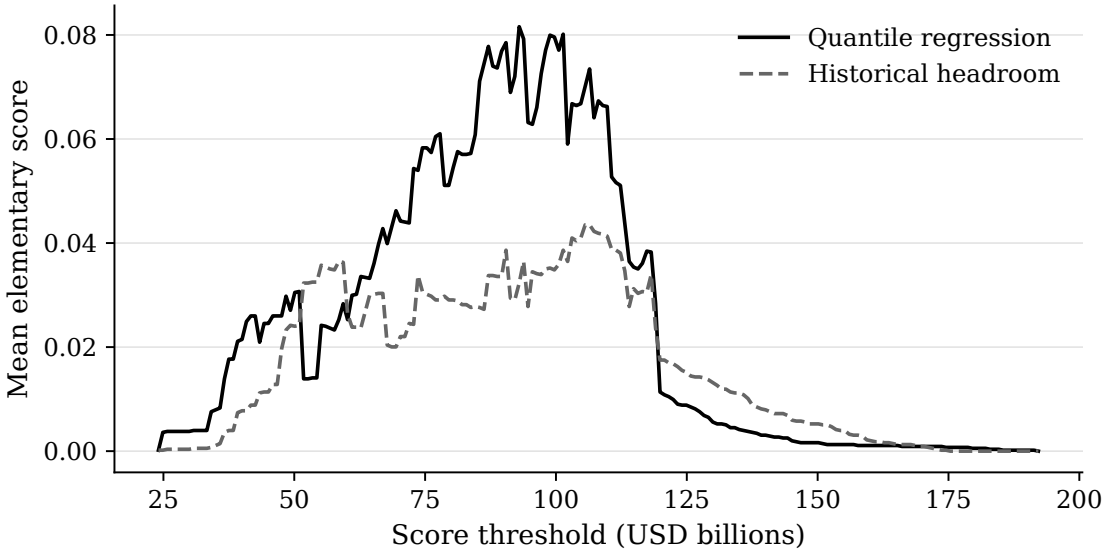


Figure 3: Murphy diagram for the original Treasury-bill primary. The saved full-breakpoint comparison also crosses. A single average-score ordering leaves other consistent-score preferences unresolved.

7.4 Controlled and within-record extensions

The controlled mean-reverting level study with GARCH innovations rejects the proposed narrow-regime explanation (Bollerslev, 1986). Each regression shows clear gains in only 2 of 12 target cells. Both also show clear gains in 6 of 60 outside cells, including homoskedastic and longer-horizon cases. The experiment uses 32 independent repetition streams with common innovations across conditions and 72 evaluation issues per repetition. Figure 4 places the saved comparisons on a common history-to-horizon coordinate. These specified processes therefore do not establish a universal history, volatility or horizon boundary.

7.5 History-length patterns and their limits

The post hoc selected coordinate is $c = W/(H + 1)$, a history-to-horizon proxy that omits coefficient count and differs from estimated effective sample size. On the discovery grid, the original regression's first cutpoint is 106.5, whereas the smaller regression's first cutpoint is 27.32. Both wide endpoint envelopes condition on the selected coordinate and grid, omit selection adjustment and allow multiple crossings for the smaller regression. These descriptions motivate a test of history sensitivity under changed innovation laws and additional histories.

The discovery cutpoints summarize a grid pooled across horizons and process settings. The original regression's first interpolation joins cells with different horizons, so this pooled display leaves a common horizon-normalized curve unidentified. Follow-up screens use a separate cutpoint for each method. Their construction is algebraically invariant to dividing both its coordinate and cutpoint by coefficient count. The follow-up evidence therefore assesses the specified history regions while leaving coefficient normalization unidentified; the diagnostic addendum derives this distinction.

The confirmation processes use Student innovations, Markov volatility switching and compound-Poisson jumps to probe the history pattern under changed innovation distributions. Simple switching and jump constructions have mature financial precedents, while these experiments remain discrete mean-reverting level models with their own specified equations (Turner et al., 1989; Merton, 1976). The complete menus and chronological selection rules stay fixed, preserv-

ing the comparison between the implemented procedures rather than asserting an oracle-risk decomposition.

The new simulation grid satisfies its prespecified short-history and long-history prediction screens across the pooled cases and reported method and process subgroups. The short-history criterion is met in 47 of 48 applicable cells, while the applicable regions remain fixed from each method's discovery cutpoint. The long-history criterion is met in 21 of 24 applicable cells, supporting the loss pattern for these procedures under the new innovation laws.

The universal simplification criterion holds in 42 of 48 paired cells, so its all-cell prediction fails and the counterexamples remain part of the finding. The Supplement reports complete screen denominators and universal-simplification failures. The diagnostic addendum displays every primary-screen exception; the current repository guide links it alongside complete candidate and repetition records. Shared simulation streams make these cell counts dependent summaries, so passing the screen supplies procedure-specific evidence rather than a population significance test.

The fixed-period FX learning curves yield partly supportive first-cutpoint evidence while exposing reversals that a single history comparison would miss. Identified crossings lie within their prespecified bands, but the shorter-horizon smaller regression is already below the historical rule at the smallest history. The compatible left-censored case remains untestable as a first-cutpoint prediction, rather than supplying another identified crossing for the overall confirmation rule. At the largest history, both longer-horizon regressions lose again against their own matched historical rule, making the relative-loss relationship visibly nonmonotone. The Supplement reports every history, breach count and nominal paired-loss diagnostic, alongside the complete prediction assessment for the specified grid.

Same-dealer daily aggregation produces predominantly historical-rule score wins. Historical headroom wins 7 of the 8 specified comparisons. The exception is the longer-history four-predictor regression at the shorter daily horizon, with about a 3.0% score reduction. It records 37 breaches versus 30 for historical headroom. Aggregation changes physical histories, horizons, update cadence, predictors, evaluation span and the sampled peak together. Source identity is held fixed, but this contrast does not identify a pure frequency effect.

An expanded fixed-parameter backcast preserves the reference-sensitive pattern but supplies no fresh independent validation. For original short-horizon FX regression, matched score falls about 12.2%, while its rolling-reference advantage is about 5.3%. Matched breaches are 416 for regression and 282 for historical headroom. The raw rolling-reference DM p-value is about 0.055 and remains descriptive. Earlier rolling-validation scoring already covered these target paths, and the smaller backcast is fully nested. Penalties were chosen later, so these overlapping windows cannot be interpreted as independent replications.

Historical comparisons remain reference-sensitive, and history length supplies a useful but incomplete guide to the gains from additional conditioning.

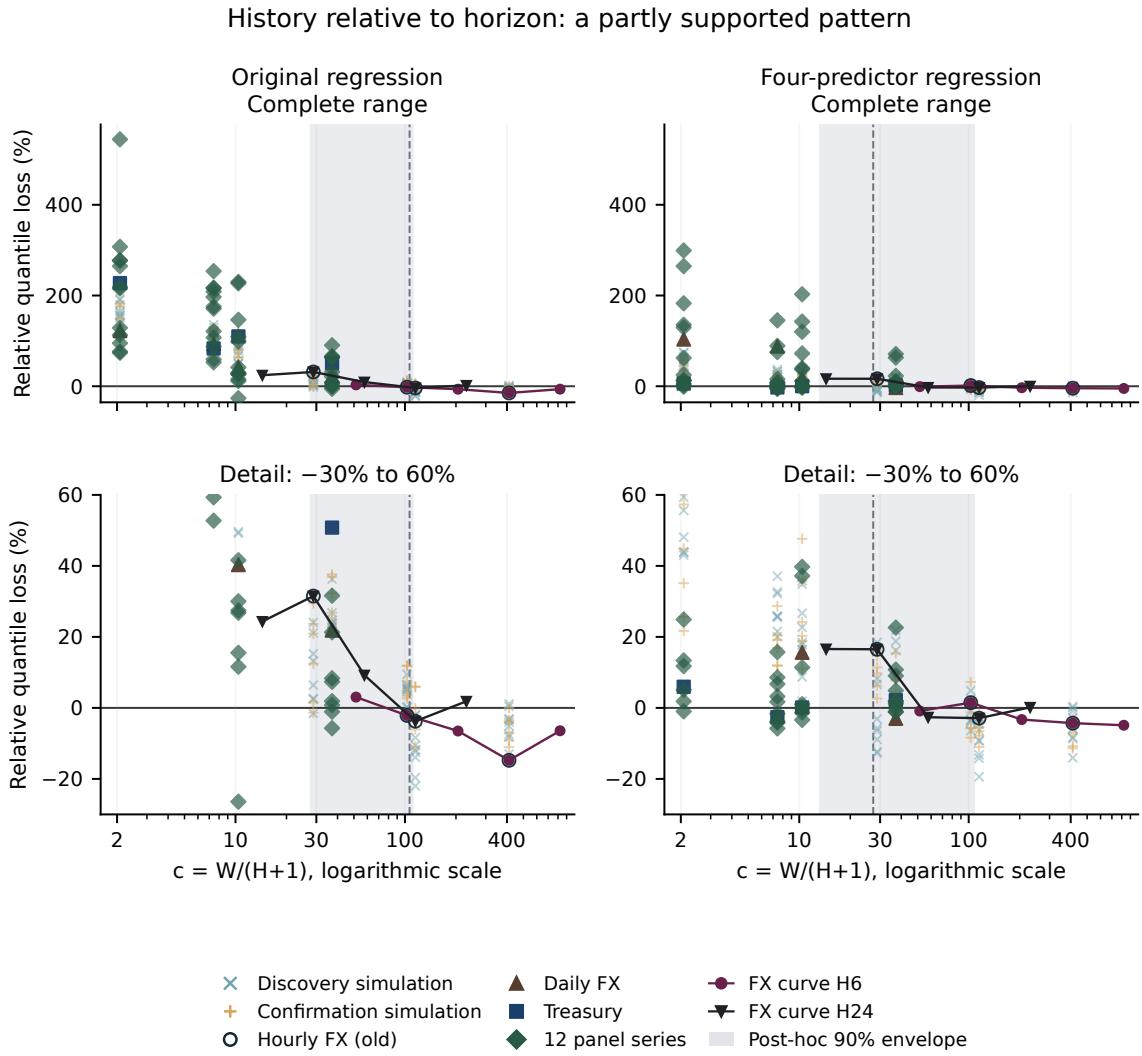


Figure 4: Conditional-regression loss against same-history historical headroom on a logarithmic $c = W/(H+1)$ axis; this coordinate excludes coefficient count. Light marks distinguish discovery and confirmation simulations; dark marks distinguish financial tasks, with lower panels repeating a labelled detail range. Discovery bands and dashed lines show post hoc first-cutpoint envelopes and point estimates from raw-coordinate interpolation across pooled horizons. Financial crossings interpolate raw history within each horizon on the reused FX period: three agree with prespecified intervals, while the remaining case stays untestable through left censoring. Both longer-horizon curves lose again at the largest history, and the universal simplification criterion fails in the confirmation experiment. Consistency supports a partial history pattern; horizon collapse and coefficient normalization remain unidentified.

8 Implications for model validation

Peak-limit validation should begin with the information set and the comparison rule before examining a candidate model's favorable performance statistic. Admit a historical response only after its whole peak horizon has matured, and construct all transformations from the eligible training observations. Then compare the candidate with a historical rule using the same history and update schedule before returning to the original operational reference. This sequence separates the matched model comparison from the combined effects of a different history length and a different update schedule.

A practical validation record should also keep scores, breaches, and stability together rather than replacing one assessment with another after seeing the results. Report the mean quantile loss together with breach frequency, overrun amounts, and unused headroom under the same issue grid. Repeat the comparison across the available periods and related series, while displaying every failed or adverse configuration beside the successful settings. Table 6 organizes observed comparison settings and the validation questions they raise, including the daily and longer-horizon exceptions.

For a limit manager, the key decision is whether extra conditioning improves the chosen monitoring objective reliably enough to justify the additional estimation. The original hourly FX result motivates further evaluation of its named regression, while the weekly and daily comparisons support demanding historical benchmarks. Calibration and combination comparisons should enter that decision through their own measured score and breach behavior, with their definitions kept explicit. The appropriate default is a matched validation process that lets comparative evidence, rather than model complexity, determine the preferred peak-limit report.

A desk must specify its monitoring objective and breach constraint before turning comparative forecast scores into a model-adoption decision. The marginal reports leave horizon reconciliation and breach responses unevaluated, so an operating policy requires its own validation.

A practical history screen is to examine the entire matched-loss curve around the available history and verify its breach behavior and temporal stability. The coordinate and its cutpoints can organize that investigation, but left censoring and later reversals require testing the actual reporting configuration. Treat a favorable region as a candidate for further validation rather than a permanent minimum-sample rule or an empirically measured sample-per-parameter threshold. The first cutpoint describes a retrospective comparison; it does not authorize switching a model on or changing its training history.

Table 6: Observed comparison settings and validation questions. Entries organize retrospective evidence rather than a causal frequency effect or deployment policy. The map guides history-sensitivity checks alongside breach behavior and operating suitability.

Frequency / history	Horizon	Evidence	Validation action
Hourly / longer history	Short	Repeated matched score gains; separate wrapper check	Compare named control and breaches
Hourly / shorter history	Short	Small score gain with more breaches	Require stable joint performance
Hourly / either history	Longer	Matched backcast gains coexist with reference-sensitive losses; coherence and preference sensitivity remain	Check the named reference, horizon order and loss preferences
Daily / same dealer	Horizons in nonempty days	Historical score wins predominate; one four-predictor gain has more breaches	Retain both controls and assess their own breaches
Weekly / short or long history	Short or longer	Frequent regression losses despite expanded penalties	Compare historical and parsimonious controls
Any source / endpoint bound	Specified horizon	Population premises and plug-in behavior differ	Check absolute displacement and dependence

9 Limitations

The FX evidence comes from a single market maker's client-flow record, limiting conclusions about other dealers, currencies, and market-making arrangements. Its original evaluation history had already been viewed, and the earlier chronological comparison reuses familiar observations from the same source. These overlapping windows and the previously viewed later period describe stability within the record rather than independent dealer replication or prospective performance. The expanded prefix includes the earlier backcast and uses penalties selected after its issue dates. A stronger external assessment requires a compatible additional dealer source and a future evaluation protocol fixed before the new outcomes become available.

The weekly data are signed position-category aggregates, so offsetting exposures across dealers can reduce the aggregate while individual gross exposures remain large. Related categories share a reporting system and economic components, which limits an interpretation of their comparisons as independent replications across institutions. The downloaded economic-date series also differs from a reconstruction of the observations available at each historical publication time. The sampled target measures recorded positions; excursions between observations, settlement cash needs, and the costs of actual limit decisions require additional measurement.

The history coordinate is descriptive, with parameter count absent from the selected index and dependence beyond the response horizon left unmeasured. Different penalty menus and approximation errors move together with predictor count, so the comparison does not isolate an estimation-cost channel. The new simulation's specified loss-pattern screens are met, while universal simplification fails and the financial grid supplies partial first-cutpoint consistency. Wide post hoc endpoint envelopes and left censoring limit threshold precision, while the longest-history reversals contradict a permanent operating cutoff. Weekly grids lack a dedicated broad-history learning curve, and some exceed the smaller model's point cutpoint without uniformly winning.

The later analyses and working-reference tests describe the observed records after earlier results had informed the research process and additional comparisons. Their p-values therefore do not account for the full sequence of model, sample, diagnostic, and presentation choices made during that process.

Filtered-path, quantile-forest and headroom-SAV comparisons evaluate finite, resource-bounded specifications rather than entire method classes. Numerical or structural failures remain unavailable comparisons; these executions neither inherit source-method guarantees nor evaluate sequential predictive conformal inference or simultaneous-trajectory calibration. The finite calibration wrapper, empirical endpoint rule, and dependent financial paths also require distinctions from the population assumptions in Proposition 2. The main evidence supports a bounded comparative forecasting result, with external validity, prospective calibration, and economic decision value remaining questions for future evaluation. The score uses a nineteen-to-one error-weight ratio for evaluation, which does not measure the actual costs of a limit decision.

10 Conclusion

Historical rules remain demanding competitors for sampled peak-position quantiles, with conditional gains depending on the reference, training history and forecast horizon. The pooled discovery display summarizes the original simulations, and the new grid supports the prespecified method-specific history regions. Coefficient normalization and a causal estimation-cost channel remain unidentified. Retrospective FX curves partly match the transferred bands, while left censoring, long-history reversals and failed universal simplification qualify a single-threshold interpretation. The controlled study's earlier narrow-regime explanation still fails, so the history pattern adds a bounded organizing description rather than replacing contrary evidence. These results motivate validation at the actual history and reference, combining score, breach frequency and stability before a limit manager adopts additional conditioning.

The weekly comparisons show that the successful FX specification does not transport reliably to related peak-position tasks at the lower frequency. Heavier regularization reduces the additional panel's median excess loss substantially, yet historical rules still win most comparisons and the conservative-selection exercise misses its success criteria. Lower average quantile loss, fewer breaches and stable rankings impose different requirements, and these data frequently rank them differently. For managers selecting peak limits, their joint assessment provides a firmer basis for action than a favorable score considered on its own.

Future work should evaluate these comparisons prospectively and obtain a compatible independent dealer record, while preserving the same response and information clock. The proposed release protocol remains unexecuted, with capture, missing observations and failed-fit continuity awaiting an implemented and evaluated issue process. A decision study could then measure the economic consequences of changing a limit, including monitoring actions, funding, hedging, and inventory adjustments. Such evidence would connect forecast quality to realized operating value under a specified policy rather than treating asymmetric error weights as observed costs. Peak-limit forecasting advances when its comparisons remain matched, its failures remain visible, and its practical claims stay tied to what the data measure.

Data and code availability

Data availability. The FX input is the specified version of the Kaggle Net Position Time Series Dataset. The owning metadata labels its license "Data files © Original Authors." This repository supplies the acquisition route and input fingerprint, but excludes raw FX observa-

tions and private per-issue prediction vectors. Obtain the source from https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset subject to the original rights holders' terms.

The New York Fed aggregates are available from its Primary Dealer Statistics service under the Bank's Terms of Use. The repository preserves the unchanged accepted Treasury snapshot with attribution and identifies study calculations separately. Copying or distributing that content remains subject to its source notices, attribution, modification labelling, continuing conditions and endorsement restrictions: https://www.newyorkfed.org/privacy/termsofuse.html. The code license does not replace those source conditions.

Code availability. Code, saved results and version-specific reproduction instructions are available at https://github.com/ByronLi0207/financial-mathematics-paper-20261020. The earlier controlled grid, same-dealer daily contrast and overlapping backcast remain complete. This revision adds the post hoc coordinate comparison, pre-run simulation screen and fixed-period financial learning curve, with all counterexamples and unavailable prediction statuses visible. The completed results include adverse, failed and unavailable cases; arithmetic readback verifies artifact consistency rather than independent scientific replication.

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References

Arjani, N., Li, F., & Sabetti, L. (2020). *Monitoring intraday liquidity risks in a real time gross settlement system*. Payments Canada. https://www.payments.ca/sites/default/files/discussion_paper_monitoring_intraday_liquidity_risks_in_a_real_time_gross_settlement_system.pdf.

Barone-Adesi, G., Giannopoulos, K., & Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, *19*(5), 583–602. https://doi.org/10.1002/(SICI)1096-9934(199908)19:5<583::AID-FUT5>3.0.CO;2-S.

Board of Governors of the Federal Reserve System (2013, March). *Reporting guidelines for preparing the FR 2004 primary government securities dealers reports*. Board of Governors of the Federal Reserve System. https://www.federalreserve.gov/apps/reportingforms/Download/DownloadAttachment?guid=f555fe11-e149-479b-8088-4ed4e9ae1206.

Board of Governors of the Federal Reserve System (2024, July). *Instructions for the preparation of government securities dealers reports: Reporting form FR 2004*. Board of Governors of the Federal Reserve System. https://www.federalreserve.gov/apps/reportingforms/Download/DownloadAttachment?guid=950ab256-b485-4586-99b0-b27eb205fdfa.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, *31*(3), 307–327. https://doi.org/10.1016/0304-4076(86)90063-1.

Boudoukh, J., Richardson, M., Stanton, R., & Whitelaw, R. F. (2004). MaxVaR: Long-horizon value at risk in a mark-to-market environment. *Journal of Investment Management*, *2*(3), 1–6. https://pages.stern.nyu.edu/~rwhitela/papers/maxvar%20joim04.pdf.

Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, *39*(4), 841–862. https://doi.org/10.2307/2527341.

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, *13*(3), 253–263. https://doi.org/10.1080/07350015.1995.10524599.

Duffie, D., Fleming, M. J., Keane, F. M., Nelson, C., Shachar, O., & Van Tassel, P. (2023). *Dealer capacity and U.S. Treasury market functionality* (Staff Report No. 1070). Federal Reserve Bank of New York Staff Reports. https://doi.org/10.59576/sr.1070.

Ehm, W., Gneiting, T., Jordan, A., & Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *78*(3), 505–562. https://doi.org/10.1111/rssb.12154.

Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, *22*(4), 367–381. https://doi.org/10.1198/073500104000000370.

Federal Reserve Bank of New York (n.d.). *Primary dealer statistics* [Data set]. Federal Reserve Bank of New York. https://www.newyorkfed.org/markets/counterparties/primary-dealers-statistics. Retrieved October 23, 2026.

Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, *74*(6), 1545–1578. https://doi.org/10.1111/j.1468-0262.2006.00718.x.

Gibbs, I., & Candès, E. J. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, *34*, 1660–1672. https://proceedings.neurips.cc/paper_files/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, *106*(494), 746–762. https://doi.org/10.1198/jasa.2011.r10138.

Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, *13*(2), 281–291. https://doi.org/10.1016/s0169-2070(96)00719-4.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70. https://www.jstor.org/stable/4615733.

Hull, J., & White, A. (1998). Incorporating volatility updating into the historical simulation method for value-at-risk. *The Journal of Risk*, *1*(1), 5–19. https://doi.org/10.21314/jor.1998.001.

Koenker, R., & Bassett, G., Jr. (1978). Regression quantiles. *Econometrica*, *46*(1), 33–50. https://doi.org/10.2307/1913643.

Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, *3*(2), 73–84. https://doi.org/10.3905/jod.1995.407942.

Lindsay, D. (2020). *Net position time series dataset* (Version 2) [Data set]. Kaggle. https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset.

McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, *7*(3–4), 271–300. https://doi.org/10.1016/s0927-5398(00)00012-8.

Meinshausen, N. (2006). Quantile regression forests. *Journal of Machine Learning Research*, *7*(35), 983–999. https://www.jmlr.org/papers/v7/meinshausen06a.html.

Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, *3*(1–2), 125–144. https://doi.org/10.1016/0304-405X(76)90022-2.

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, *55*(3), 703–708. https://doi.org/10.2307/1913610.

Petrunia, R., Sabetti, L., & Voia, M. (2019). *Tail risk in the automated clearing settlement system (ACSS)* (Discussion Paper No. 7). Payments Canada. https://www.payments.ca/sites/default/files/tail_risk_in_the_automated_clearing_settlement_system_acss.pdf.

Sussman, E. (2023). A strengthened Orlicz–Pettis theorem via Itô–Nisio. *Annals of Functional Analysis*, *14*(1), Article 22. https://doi.org/10.1007/s43034-022-00225-1.

Turner, C. M., Startz, R., & Nelson, C. R. (1989). A Markov model of heteroskedasticity, risk, and learning in the stock market. *Journal of Financial Economics*, *25*(1), 3–22. https://doi.org/10.1016/0304-405X(89)90094-9.

van den Hoogen, S. (2017). *Intraday liquidity risk estimation using transaction data: An extreme value theory approach* [Master's thesis, Delft University of Technology]. https://resolver.tudelft.nl/uuid:1c0c4545-bb07-4916-aa37-1d4bea3b7872.

Viering, T., & Loog, M. (2023). The shape of learning curves: A review. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(6), 7799–7819. https://doi.org/10.1109/TPAMI.2022.3220744.

Wisniewski, W. J. (2024). *Net position and retail trader behaviour: Implications for market makers' risk management strategies* [Doctoral thesis, Royal Holloway, University of London]. https://pure.royalholloway.ac.uk/en/publications/net-position-and-retail-trader-behaviour-implications-for-market-/.

Wisniewski, W., Lindsay, D., & Lindsay, S. (2020). Application of conformal prediction interval estimations to market makers' net positions. In *Proceedings of the Ninth Symposium on Conformal and Probabilistic Prediction and Applications* (Proceedings of Machine Learning Research, Vol. 128, pp. 285–301). https://proceedings.mlr.press/v128/wisniewski20a.html.